

SHA TIN, Hong Kong

WMO: 450390

Lat: 22.4025N

Lon: 114.2100E

Elev: 42

StdP: 14.67

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 46.0	(c) 48.4	(d) 24.5	(e) 18.7	(f) 55.6	(g) 29.9	(h) 24.0	(i) 55.5	(j) 12.0	(k) 56.4	(l) 11.1	(m) 57.8	(n) 5.5	(o) 340	(p) 0.304

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.1	(c) 93.0	(d) 79.7	(e) 91.5	(f) 79.7	(g) 90.2	(h) 79.4	(i) 81.9	(j) 88.7	(k) 81.4	(l) 88.0	(m) 80.9	(n) 87.2	(o) 6.3	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.1	156.7	86.1	79.6	154.0	85.6	79.1	151.2	85.1	46.0	89.3	45.3	88.0	44.7	87.4	90.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.5	(b) 11.8	(c) 10.4	(e) 42.4	(f) 96.9	(g) 4.3	(h) 1.3	(i) 39.4	(j) 97.9	(k) 36.9	(l) 98.7	(m) 34.5	(n) 99.4	(o) 31.4	(p) 100.4		
(a) 13.5	(b) 11.8	(c) 10.4	(e) 36.1	(f) 85.4	(g) 3.9	(h) 2.7	(i) 33.3	(j) 87.3	(k) 31.0	(l) 88.9	(m) 28.9	(n) 90.4	(o) 26.0	(p) 92.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	74.4	60.6	62.7	66.7	73.4	79.4	83.3	84.7	84.3	83.0	78.4	71.7	63.6
	DBStd	9.82	5.70	6.89	5.81	5.01	4.01	2.87	2.44	2.55	2.65	3.44	5.09	5.53
	HDD50	8	4	3	0	0	0	0	0	0	0	0	0	1
	HDD65	424	156	111	50	5	0	0	0	0	0	0	11	91
	CDD50	8904	332	359	517	701	912	998	1075	1063	991	881	651	424
	CDD65	3843	19	46	102	256	447	548	610	598	541	416	211	49
	CDH74	37822	26	99	269	1340	4077	6463	7740	7390	6206	3313	828	71
	CDH80	13575	0	2	16	224	1186	2511	3419	3117	2294	729	75	2
Wind	WSAvg	5.3	5.0	5.2	5.3	5.4	5.3	6.0	6.0	5.2	5.0	5.1	5.1	5.1
Precipitation	PrecAvg	75.7	1.1	1.5	2.1	5.1	10.1	16.0	12.7	12.6	9.3	2.7	1.4	1.1
	PrecMax	113.2	9.7	6.7	8.2	23.1	22.2	52.9	39.6	32.9	20.9	16.4	10.0	7.5
	PrecMin	43.5	0.0	0.0	0.2	0.0	1.6	3.8	3.0	1.2	2.0	0.0	0.0	0.0
	PrecStd	18.9	1.8	1.9	2.0	4.7	4.7	10.3	8.0	9.1	5.5	4.0	2.1	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.2	79.7	81.9	86.6	91.7	92.0	95.9	95.6	94.2	90.0	84.7	79.0
		MCWB	66.6	70.9	72.7	76.2	79.3	79.6	81.0	80.1	79.0	76.6	73.1	67.9
	2%	DB	73.4	77.1	79.5	84.3	89.3	90.7	92.4	93.0	91.4	87.5	82.1	76.0
		MCWB	65.0	70.1	71.7	75.6	79.1	79.8	80.2	79.9	79.1	75.5	71.7	66.0
	5%	DB	70.2	74.6	77.2	82.6	87.4	89.6	91.0	91.2	89.7	85.5	80.0	73.4
		MCWB	62.7	68.1	70.8	74.9	78.6	79.8	79.9	79.9	78.1	74.2	70.9	64.5
	10%	DB	68.0	71.7	74.4	80.8	85.5	88.1	89.7	89.6	88.2	83.9	78.5	71.1
		MCWB	61.7	66.3	68.6	74.3	78.1	79.7	79.8	79.6	77.9	73.8	70.6	63.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.0	73.4	74.5	78.3	81.0	82.2	83.5	83.2	81.8	80.3	75.8	70.9
		MCDB	74.1	78.1	79.3	83.8	87.2	88.0	89.1	91.1	88.7	85.7	81.6	75.2
	2%	WB	66.6	71.4	73.2	77.3	80.3	81.6	82.0	81.9	80.8	78.5	74.7	68.8
		MCDB	71.3	75.5	78.5	82.3	86.2	87.3	89.0	89.0	87.6	83.8	79.8	73.0
	5%	WB	64.8	69.2	71.8	76.3	79.7	81.0	81.1	81.2	80.0	77.3	73.6	67.0
		MCDB	68.8	73.4	75.9	81.1	85.6	86.7	87.8	88.0	86.6	83.2	78.3	71.9
	10%	WB	63.0	67.1	70.1	75.1	79.0	80.5	80.6	80.5	79.3	76.2	72.4	64.8
		MCDB	66.9	70.5	73.3	79.9	84.8	86.2	87.1	87.0	85.5	82.0	76.9	69.9
Mean Daily Temperature Range	5% DB	MDBR	10.2	9.4	9.2	8.8	8.3	7.9	9.1	9.6	9.5	9.6	9.6	10.5
		MCDBR	12.8	13.3	13.2	11.8	10.8	9.7	11.2	12.3	11.6	11.4	11.1	12.9
		MCWBR	6.1	6.7	6.6	5.3	3.8	3.3	3.7	4.3	4.0	4.6	4.8	6.0
	5% WB	MCDBR	10.6	10.7	11.1	10.0	9.2	8.1	9.8	10.6	10.5	9.0	8.8	10.5
		MCWBR	6.2	6.2	6.4	5.4	3.9	3.3	4.0	4.4	4.0	4.3	4.5	6.1
Clear-Sky Solar Irradiance	taub	0.582	0.635	0.763	0.740	0.578	0.547	0.535	0.600	0.633	0.676	0.575	0.568	
	taud	1.800	1.727	1.519	1.601	1.988	2.087	2.116	1.917	1.820	1.680	1.882	1.855	
	Ebn at Noon	215	214	194	203	238	243	246	231	220	201	216	213	
	Edh at Noon	63	72	92	86	58	52	50	62	67	74	57	58	
All-Sky Solar Radiation	RadAvg	973	989	1066	1186	1420	1456	1695	1589	1498	1394	1129	1013	
	RadStd	159	157	144	181	193	219	179	146	112	140	128	149	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	+0.45	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page